



- 6 The discrete random variable X has probability generating function $G_X(t)$ given by

$$G_X(t) = \frac{t}{(3-2t)^2}.$$

(a) Find $E(X)$ and $\text{Var}(X)$.

[5]

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The discrete random variable Y has probability generating function $G_Y(t)$ given by

$$G_Y(t) = \frac{t^2}{(3-2t)^2}.$$

The random variable Z is the sum of the random variables X and Y .

(b) Assuming X and Y are independent, find $P(Z > 4)$. [5]

This image shows a full page of a handwriting practice worksheet. It consists of multiple sets of three horizontal dashed lines, providing a guide for letter height and placement. The lines are evenly spaced across the entire page, leaving ample room for writing practice. There is no text or other markings on the page.